

Reviewer Pack — Minimal Rank of Noncommutative Syzygies over DPLL Search Tre...

Entry f8ed65bf56e7 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-26 14:58:52 UTC

Minimal Rank of Noncommutative Syzygies over DPLL Search Trees

Entry ID: f8ed65bf56e7 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 14:58:52 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry (Syzygies) **Field B** (complexity object): Complexity Theory: DPLL Search Tree Width

Statement:

[‘For every Boolean formula F with n variables, the minimal rank of its noncommutative syzygetic complex $\rho(F)$ is at least $\Omega(n^2 \log n)$, where the logarithm is base 2.’, ‘Equivalently, for any constant $c > 0$, there exists a family of Boolean formulas $\{F_n\}$ such that $\rho(F_n) \geq cn^2 \log^2 n$ for all n .’]

Rationale (proposer’s reasoning):

[‘Noncommutative syzygies have been used in algebraic geometry to study the complexity of linear spaces. Their relationship with DPLL search trees, which represent the decision-making process in SAT solvers, could potentially provide insights into the hardness of SAT problems.’, ‘The conjecture is motivated by the fact that noncommutative algebras capture certain aspects of quantum information theory, and it is plausible that these structures can be exploited to analyze complexity classes.’]

Taxonomy category: GCT_DET_PERM (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b08923798d3ba9d2

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all $n \leq 40$, the minimal rank of the noncommutative syzygetic complex $\rho(F_n)$ is at least $\Omega(n^2 \log^2 n)$, with a confidence interval of 95% based on an aggregate mean across all seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - noncommutative geometry AND syzygies AND DPLL search tree width - syzygetic complex rank lower bound ON Boolean formulas AND complexity theory - $\Omega(n^2 \log n)$ minimal rank in noncommutative geometry FOR DPLL tree widths

Top relevant hits considered: - [http://arxiv.org/abs/math/0610043v5] Lecture Notes on Non-commutative Algebraic Geometry and Noncommutative Tori - [http://arxiv.org/abs/0705.1265v2] A noncommutative Bohnenblust-Spitzer identity for Rota-Baxter algebras solves Bogoliubov's recursion - [http://arxiv.org/abs/1411.6751v4] The noncommutative infinitesimal equivariant index formula: part II - [http://arxiv.org/abs/1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [http://arxiv.org/abs/cs/0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [http://arxiv.org/abs/0908.4453v1] Depth-Independent Lower bounds on the Communication Complexity of Read-Once Boolean Formulas - [http://arxiv.org/abs/2510.17487v1] Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observin - [http://arxiv.org/abs/1411.4413v2] Observation of the rare $B_s^0 \rightarrow \mu^+ \mu^-$ decay from the combined analysis of CMS and LHCb data

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def frege_proof_width(formula):
    if formula[0] != '(':
        return 1
    stack = []
    for char in formula:
        if char == '(':
            stack.append(char)
        elif char == ')':
            stack.pop()
    return len(stack)

def generate_random_formula(n):
    variables = [f'x{i}' for i in range(1, n+1)]
    clauses = []
    for _ in range(n):
```

```

        clause = random.sample(variables, 2)
        clause.append(random.choice(['|', '&']))
        clauses.append(''.join(clause))
    return '(' + ' '.join(clauses) + ')'

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    total_rank = 0
    instances_tested = sum(n_values)

    for n in n_values:
        formula = generate_random_formula(n)
        rank = frege_proof_width(formula)
        total_rank += rank

    mean_value = total_rank / instances_tested
    conjecture_holds = mean_value >= n**2 * math.log2(n) * 1.5
    counterexample = "" if conjecture_holds else f"rank={mean_value}, expected={n**2 * math.log2(n)}

    return {
        "metric_name": "frege_proof_width",
        "metric_value": mean_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]]
    if not seeds:
        from sympy.ntheory import primerange
        seeds = list(primerange(2, 30))

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result['metric_value'] for result in results) / len(results)
    support_fraction = sum(1 for result in results if result['conjecture_holds']) / len(results)

    if all(result['conjecture_holds'] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction:.2f}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result['conjecture_holds'])
        print(f"RESULT: FALSIFIED counterexample={result['counterexample']} first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
' : 'rank=0.0, expected=8515.084951819781'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}
TRIAL: {'metric_name': 'frege_proof_width', 'metric_value': 0.0, 'instances_tested': 120, 'conjecture': 'The rank of the noncommutative syzygetic complex  $\rho(F)$  is significantly lower than the expected value, indicating that the conjecture is falsified.'}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c373161.py", line 83, in <module>
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")
    ^^^^^^
NameError: name 'result' is not defined. Did you mean: 'results'?
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results show that for all trials, the rank of the noncommutative syzygetic complex $\rho(F)$ is significantly lower than the expected value, indicating that the conjecture is falsified. | next: Investigate the cause of the discrepancy between the expected and actual ranks. Consider revising the conjecture or exploring alternative methods to prove it.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11952
2	propose	ollama_remote	glm4:latest	0	0	9881
3	preregistration	ollama_remote	glm4:latest	0	0	6080
4	novelty	ollama_remote	glm4:latest	0	0	4782
5	novelty	ollama_remote	glm4:latest	0	0	6604
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11388
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7265
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14320
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8366
10	judge	ollama_remote	glm4:latest	0	0	11297

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 91935 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/f8ed65bf56e7.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/f8ed65bf56e7.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/f8ed65bf56e7.tar.gz (if generated)